Cofinal family

Lean (a, m) lives; the $4 \cdot 10^5$ table was killing the frame-A 2^k ladder

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Abstract

r456 retyped the frequently-mincut as the full-cap cone `frequently_selected_augDualResolvent_ge_half`. The named 2^k frame-A ladder dies past $k = 10$ on the campaign table (TABLE_CAP = $4 \cdot 10^5$). A sealed 76-window scan shows a sharp wall at $a = 631/641$ — exactly $\sqrt{\text{TABLE\_CAP}}$. Completing the comb to a^2 revives k_z197 (2^{10}) and generic k_z230 (exact dps $= 50$, $n_{\text{flip}} = 0$). The Lean selected sequence $a_k = 2^k$, $m_k = k \cdot 2^{\lfloor \sqrt{k} \rfloor} - 1$ is living at the cap through $k = 16$ ($k = 10$ unchanged under the full a^2 comb). One residual deep death remains: k_z137 ($N = 8300$) still flips after the comb is completed. No RH claim.

Status. No new SATZ. Ausgang `FAMILY_CANDIDATE / TABLE_CAP_ARTEFACT / RESIDUAL_DEEP_DEATH`. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.

1 Selected requirements

Proposition 1 (Lean-admissible family)**.** *From `RH/Selected.lean`:*

1. *$W^{\mathbb{R}}(a, m) = \text{ExactFold}(a, m)$ is total for every prime-power a and every mesh m (ll. 189–191, 203–205).*
2. *Cap is $(S + 1)/2$ of the folded window (l. 246). The μ-ONB uses $Q^T(\text{combHankel})\, Q = I$ (l. 287) — the full comb, not a prefix.*
3. *Coverage (`selected_covers`, ll. 412–418): one sequence with $a_k \to \infty$ and $m_k \to \infty$ supplies $a_k \ge a_0(f)$ and $m_k \ge \text{meshExp}\, f$ for every f.*
4. *The named sequence (ll. 36–41, 298–307) is $a_k = 2^k$, $r_k = \lfloor \sqrt{k} \rfloor$, $m_k = k \cdot 2^{r_k} - 1$, $\Delta_k = 2^{-r_k} \log 2$.*

Freedom: any prime-power anchors with both coordinates cofinal. The Python k_z raster couples m to the local log-gap and truncates the comb at $4 \cdot 10^5$; that is a builder choice, not a Lean constraint. r329 (EXT3) tested a different predicate (F_A/cubic) on fresh small-gap anchors and found violations; it did not test $q^\dagger$ at the cap. Those EXT3 windows are ABD-living on the capped table ($a \le 601$).

2 Capped census

Proposition 2 (sealed grid, a-wall)**.** *`SCAN_KZ` (76 indices) was frozen before the first pack. 55 live, 21 dead. Every live window has $a \le 631$; every dead window has $a \ge 641$. Holdout (every 4th, 19 windows): the pre-registered predictor $a \le 631$ scores 19/19. $N \le 1641$ scores only 14/19 (k_z69, $N = 5690$, lives). The wall equals the table: $631^2 = 398161 < 4 \cdot 10^5 < 641^2 = 410881$.*

3 Full comb

Proposition 3 (table-cap artefact)**.** *Raising the von Mangoldt table past a^2 and keeping the same (a, M, L, Δ):*

k_z	a	capped	full comb	exact
136	*631*	*live*	*live*	—
138	*643*	*dead* 4161	*live*	—
139	*647*	*dead* 2796	*live*	—
197	1024	*dead* 3788	*live*	—
230	1259	*dead* 1818	*live*	$n_{\text{flip}} = 0$ *dps*50
137	641	*dead* 7511	*dead* 7308	—

Lean $k = 10$ ($m = 79$) has the same $q^\dagger = 0.9319618718590412$ at $\text{ka} = 34034$ (capped) and $\text{ka} = 82267$ (full a^2). The r447/r448 deaths of k_z197 and k_z230 were exact arithmetic on a truncated comb, not on `ExactPrimeSource`(a).

4 Candidate

Proposition 4 (FAMILY_CANDIDATE)**.** *Rule (Lean-admissible):*

$$a_k = 2^k, \qquad m_k = k \cdot 2^{\lfloor \sqrt{k} \rfloor} - 1,$$

comb = all prime powers $\leq a^2$ (no $4{\cdot}10^5$ cut). Certified living at the cap: $k = 5, 6, 7, 8, 9, 10, 11, 12, 14, 16$ (float); $k = 10$ full-comb stable. Generic $a = 1259$ (not 2^k) is exact-living at the frame-A mesh once the comb is complete. Residual: k_z137 ($N = 8300$) still dies with a complete comb — frame-A cofinality at huge N stays open. The $a \leq 631$ predictor is a table-cap artefact and is not a death law.

5 Claim boundary

Finite-window census. Not a theorem of the Riemann hypothesis, nor evidence against it. No L^* claim. No $R^\dagger$ claim.